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Building Trees from Traversals

CS 201: Data Structures and Algorithms

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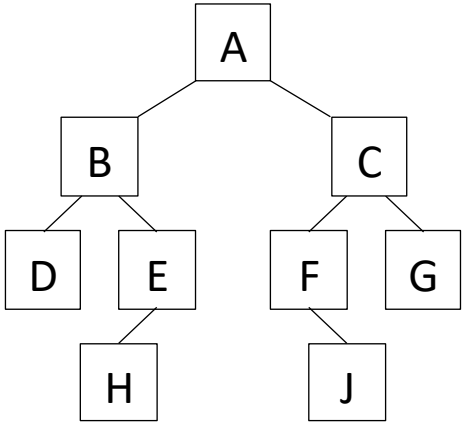
High Level View of Traversals

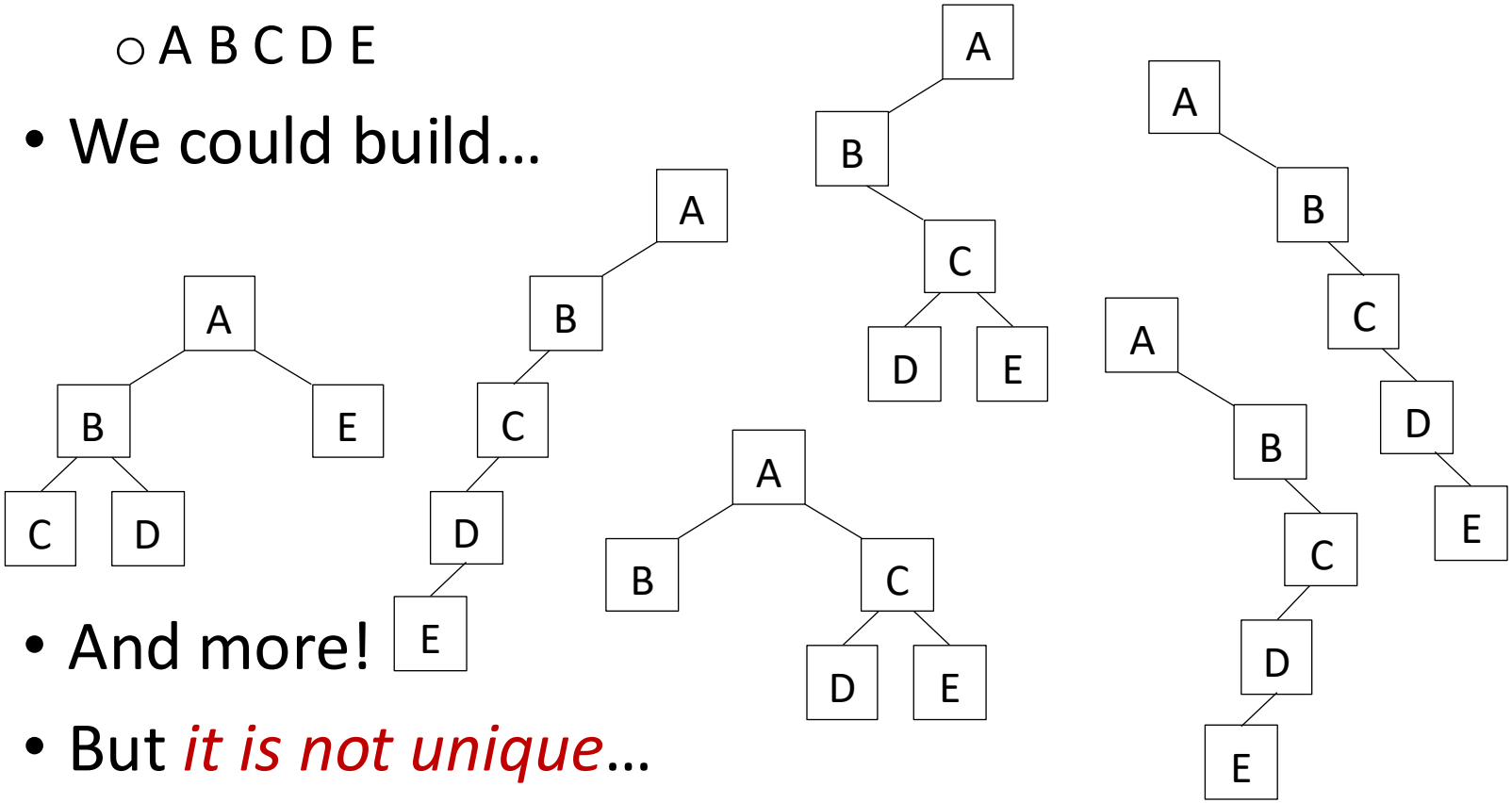
- Consider the following tree:
  - Preorder: A <u>B D E H</u> <u>C F J G</u>
  - Postorder: <u>D H E B</u> <u>J F G C</u> A
  - Inorder: <u>D B H E</u> A <u>F J C G</u>
  - Level-order: A <u>B C</u> <u>D E F G</u> <u>H J</u>



- Where is the root in each traversal?
  - Preorder: First
  - Postorder: Last
  - Inorder: Between the left and right subtrees
  - Level-order: First

Building a Tree from a Traversal

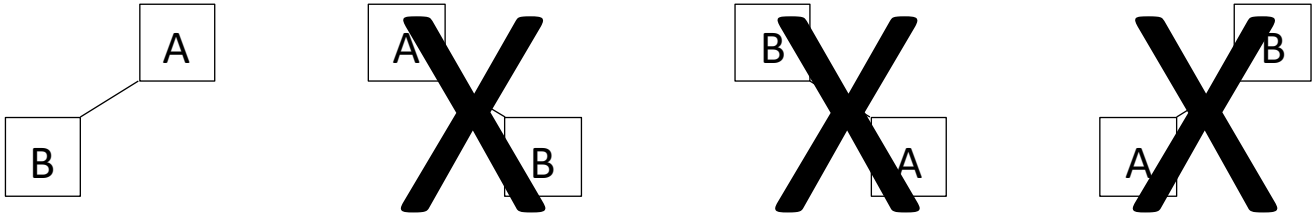
- Consider a tree with this preorder traversal:
  - A B C D E
- We could build…



- And more!
- But *it is not unique*…

Building Unique Trees

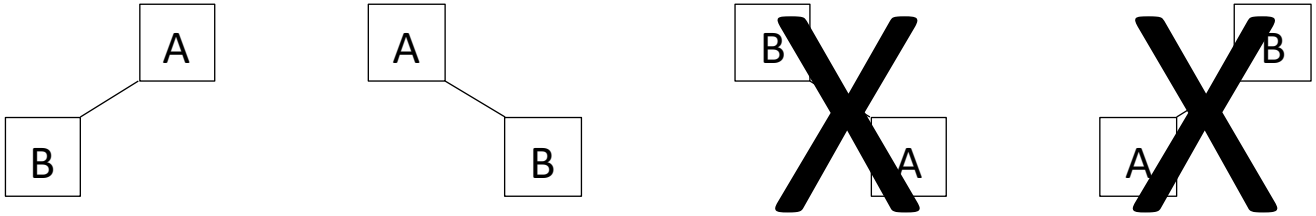
- Can we make a unique tree with all the traversals?
  - Preorder: A B
  - Postorder: B A
  - Level-order: A B
  - Inorder: B A
- Possibilities:



Building Unique Trees

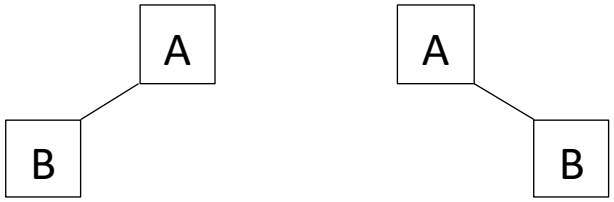
- Can we make a unique tree with these traversals?
  - Preorder: A B
  - Postorder: B A
  - Level-order: A B

- Possibilities:



Building Unique Trees

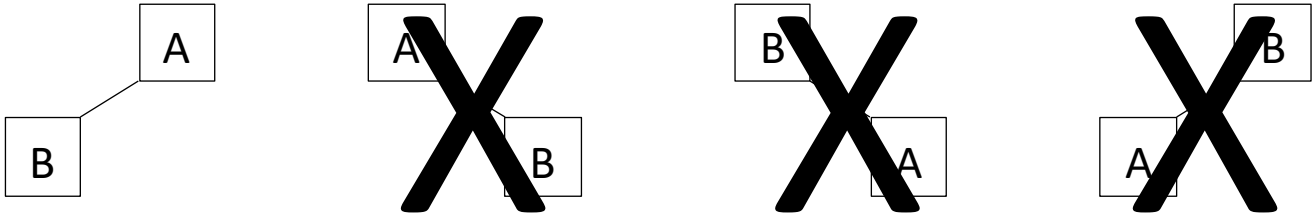
- Can we make a unique tree with these traversals?
  - Preorder: A B
  - Postorder: B A
  - Level-order: A B



- The distinguishing traversal is the inorder traversal
  - Left's Inorder: B A
  - Right's Inorder A B

Building Unique Trees

- Only the first traversal and inorder give us new info
  - **Preorder: A B**
  - Postorder: B A
  - Level-order: A B
  - **Inorder: B A**
- Possibilities:

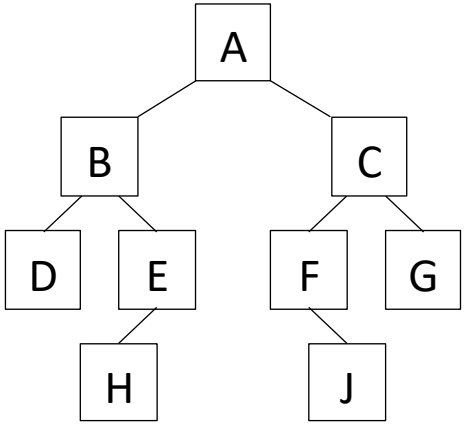


Building Unique Trees

- We can make a tree with one or more traversals
- But to guarantee a **unique** tree, we must have:
  1. Inorder Traversal
  2. Another Traversal

High Level View of Traversals

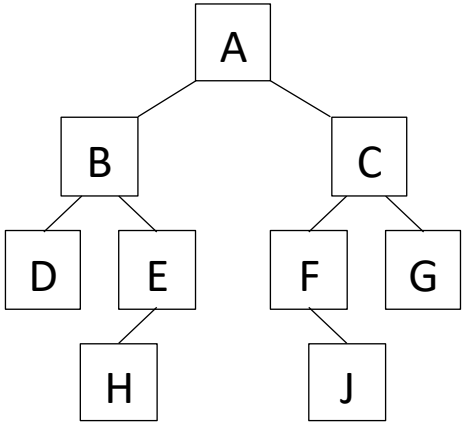
- Consider the following tree:
  - Preorder: A <u>B D E H</u> <u>C F J G</u>
  - Postorder: <u>D H E B</u> <u>J F G C</u> A
  - Inorder: <u>D B H E</u> A <u>F J C G</u>
  - Level-order: A <u>B C</u> <u>D E F G</u> <u>H J</u>



- Where is the root in each traversal?
  - Preorder: First
  - Postorder: Last
  - Inorder: Between the left and right subtrees
  - Level-order: First

High Level View of Pre and In

- Consider the following tree:
  - Preorder: A <u>B D E H</u> <u>C F J G</u>
  - Inorder: <u>D B H E</u> A <u>F J C G</u>
- Where is the root in each traversal?
  - Preorder: First
  - Inorder: Between the left and right subtrees



Tree from Preorder and Inorder

- Preorder: A B D E H C F J G
- Inorder: D B H E A F J C G

1. Identify the root

Tree from Preorder and Inorder

- Preorder: A B D E H C F J G
- Inorder: D B H E A F J C G

A

1. Identify the root
2. Add it to the tree
3. The left subtree elements are to the left of the root in the inorder traversal
4. The right subtree elements are to the right of the root in the inorder traversal

Tree from Preorder and Inorder

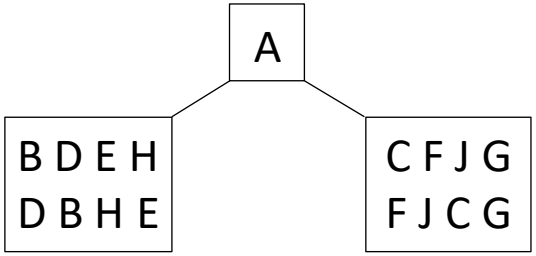
- Preorder: A B D E H C F J G
- Inorder: <u>D B H E</u> A <u>F J C G</u>

A

1. Identify the root
2. Add it to the tree
3. The left subtree elements are to the left of the root in the inorder traversal
4. The right subtree elements are to the right of the root in the inorder traversal
5. Split the data in preorder

Tree from Preorder and Inorder

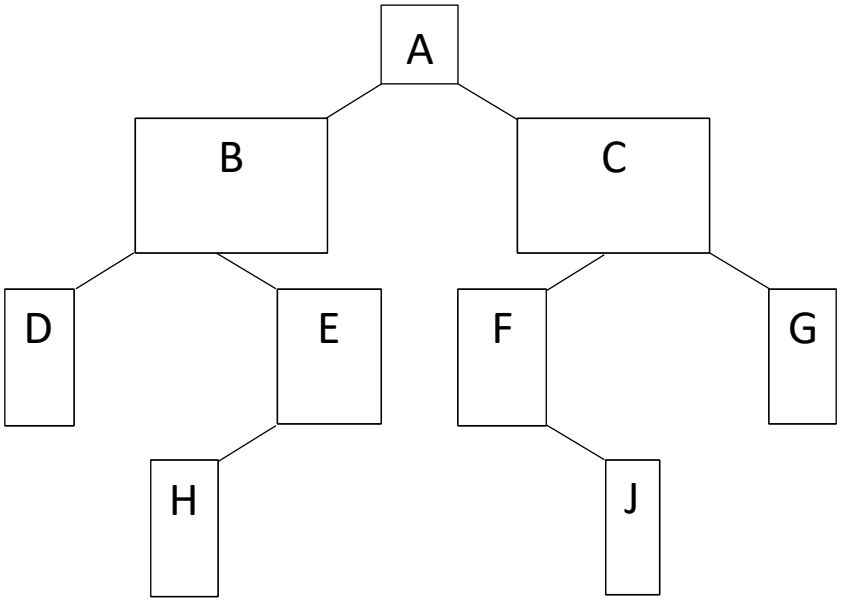
- Preorder: A <u>B D E H</u> <u>C F J G</u>
- Inorder: <u>D B H E</u> A <u>F J C G</u>



1. Identify the root
2. Add it to the tree
3. The left subtree elements are to the left of the root in the inorder traversal
4. The right subtree elements are to the right of the root in the inorder traversal
5. Split the data in preorder
6. Recurse on the left data to form the left subtree
7. Recurse on the right data to form the right subtree

Tree from Preorder and Inorder

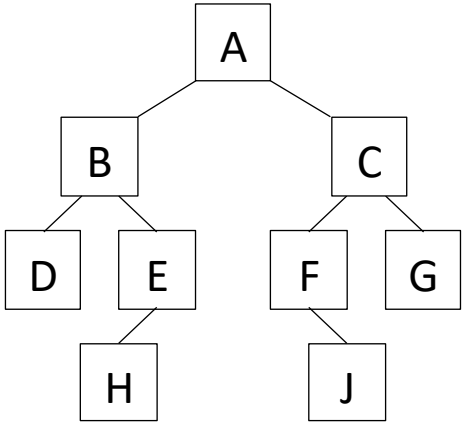
Preorder
Inorder



1. Identify the root and add it to the tree
2. Split the data
3. Recurse on the left and right data

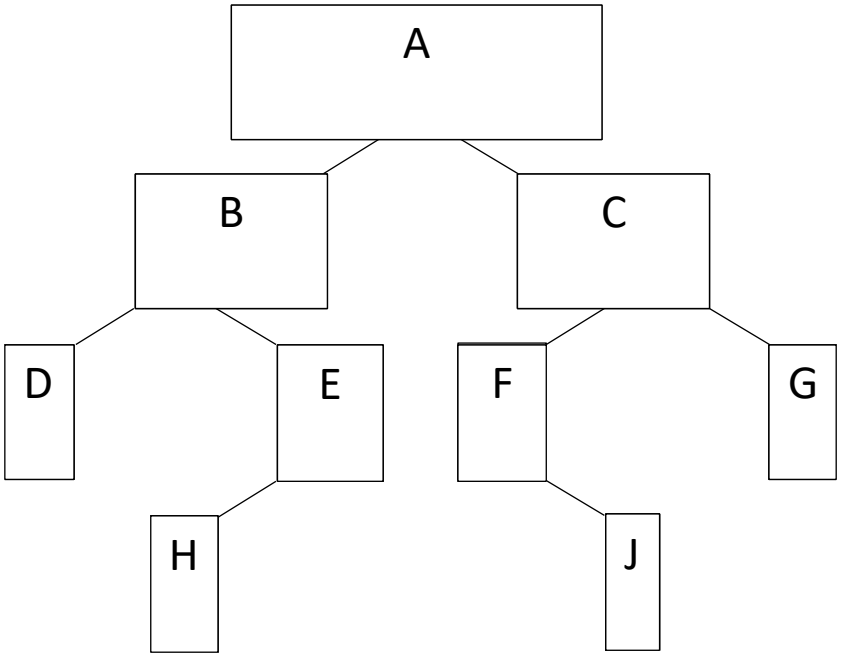
High Level View of Post and In

- Consider the following tree:
  - Postorder: <u>D H E B</u> <u>J F G C</u> A
  - Inorder: <u>D B H E</u> A <u>F J C G</u>
- Where is the root in each traversal?
  - Postorder: Last
  - Inorder: Between the left and right subtrees



Tree from Postorder and Inorder

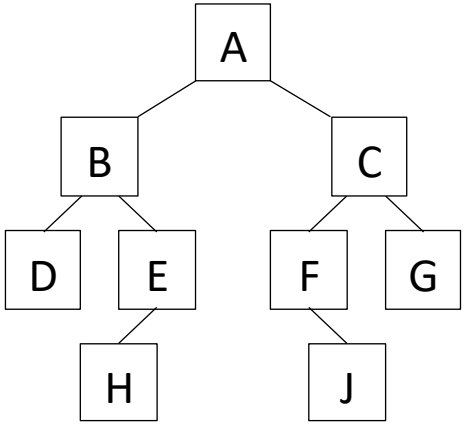
Postorder
Inorder



1. Identify the root and add it to the tree
2. Split the data
3. Recurse on the left and right data

High Level View of Level and In

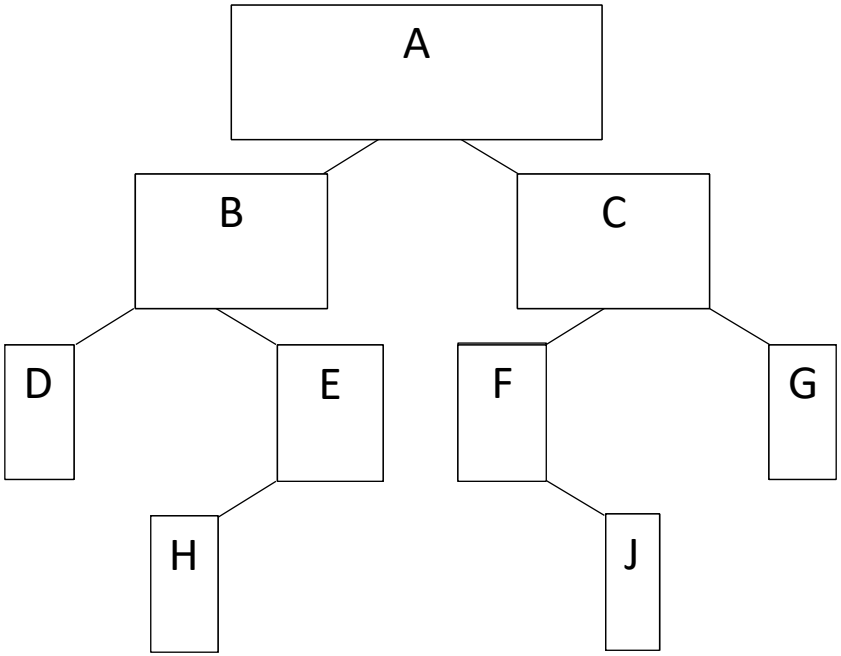
- Consider the following tree:
  - Level-order: A <u>B C</u> <u>D E F G</u> <u>H J</u>
  - Inorder: <u>D B H E</u> A <u>F J C G</u>
- Where is the root in each traversal?
  - Level-order: First
  - Inorder: Between the left and right subtrees



- The split
  - Level-order: A B C D E F G H J
  - Inorder: <u>D B H E</u> A <u>F J C G</u>
  - Level-order splits: B D E H and C F G J (keep ordering)

Tree from Level-order and Inorder

Level-order
Inorder



1. Identify the root and add it to the tree
2. Split the data
3. Recurse on the left and right data

Questions?